Computational neuroscience

Key ideas and main analysis techniques

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Computational application
to cognitive neuroscience



UNIVERSITÀ
DI TORINO

Index

Chapter 1
The brain as a network

Chapter 2
The brain as a dynamical system

Chapter 1
Brain connectivity

The idea of a connected brain

has a very long story, rooted in neuropsychological studies

Disconnection syndromes

a brain lesion can impair a function by disrupting interactions, not only by destroying a region that "contains" that function.

Wernicke - 1874
Lichteim - 1885

Aphasia

Gazzaniga -1967

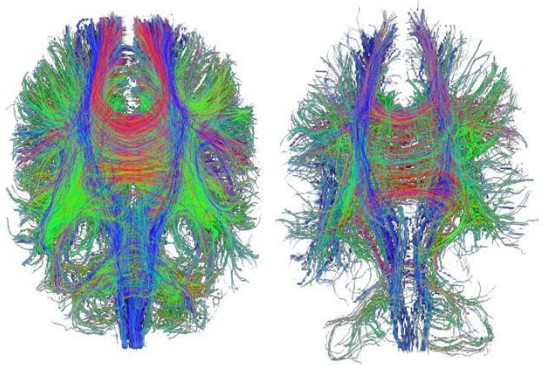
Split brain

Geschwind -1965

"Disconnexion Syndromes in Animals and Man"

Many types of connectivity

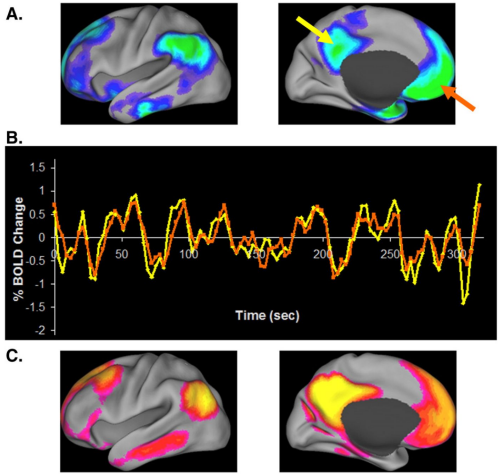
Structural



https://www.umimri.com/dti/

connections between neurons are too “slow” to explain cognitive functions…

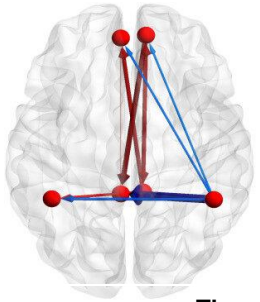
Functional



Raichle, 2007

To study the functional connectivity allows to looks at the emergent properties of the brain

Effective



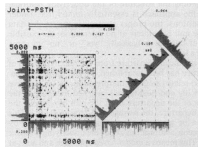
Ting, 2014

Models of causal relations between brain regions or group of neurons

Granger causality

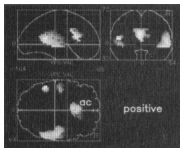
Whole brain computational modeling (e.g. Hopf model)

A brief timeline of studies about functional connectivity



Correlation of spike trains

1989
Aersten et al



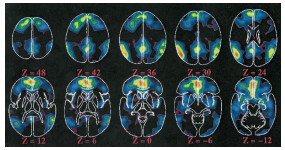
PCA-based FC with PET

1993
Frinston et al.



Seed-based FC with rs-fMRI between motor cortices

1995
Biswal et al.

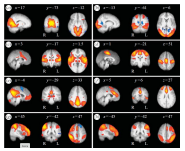


Default-mode network in PET data

2001
Raichle et al

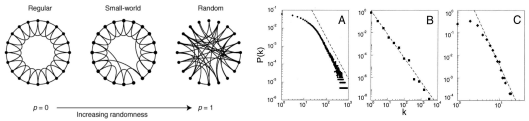
Default-mode network in fMRI data

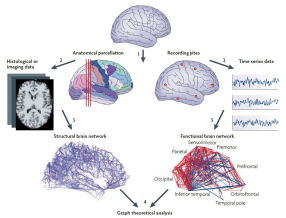
2003
Greicius et al



Resting-state networks

2005
Beckmann et al





Network Theory

1998
Watts & Strogatz

Collective dynamics of small-world networks

1999
Albert & Barabasi

Emergence of scaling in random networks

2004
Sporns et al.

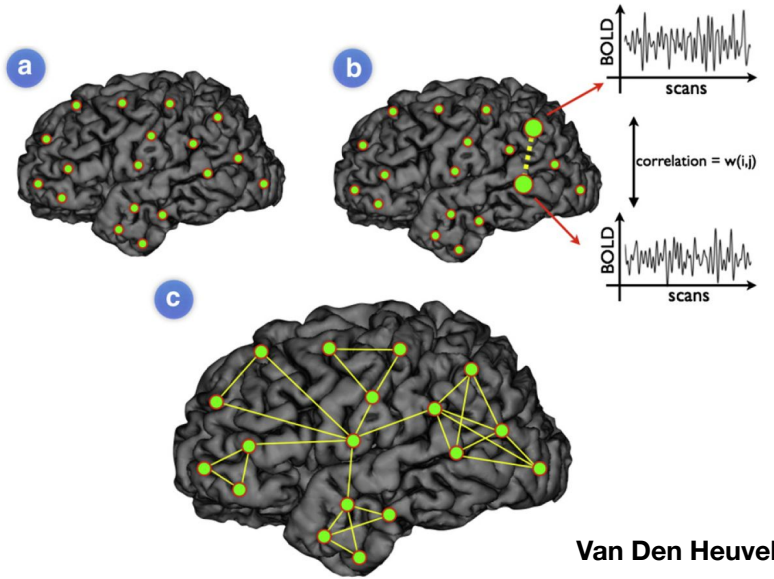
Organization of complex brain networks

2009
Bullmore & Sporns

Graph-based analysis of complex brain networks

What does Functional Connectivity (FC) mean?

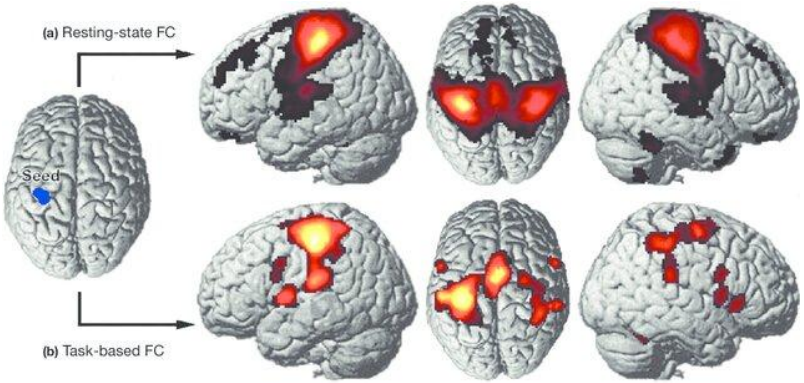
correlation of some kind of neurophysiological signal measured in different brain regions



Van Den Heuvel et al., 2010

Different approaches to FC with fMRI

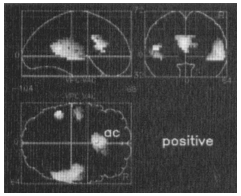
model-dependent
(seed methods)



model-free
(without a priori information)

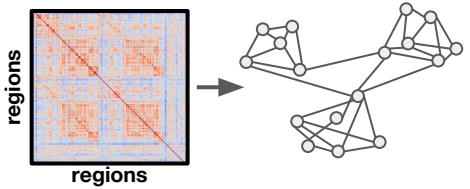
PCA

ICA



clustering methods

network-based analysis



The connectivity is often studied at rest

The incredible amount of information in the activity at rest of the brain at rest was discovered studying the "physiological noise" of the brain (cardiac and respiratory -related signal).

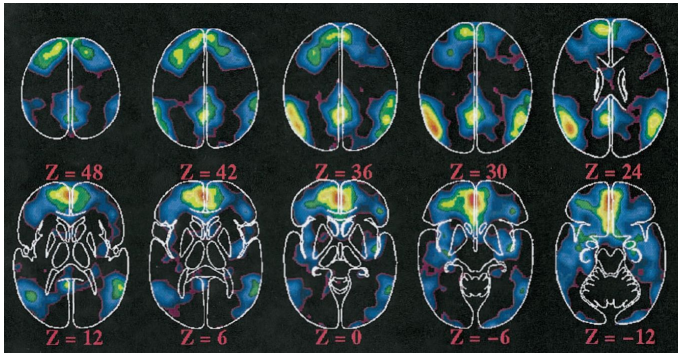
Even removing this kind of noise, some residual spatio-temporal correlation of the signal remains…

Lowe, 2010

The Default-Mode Network

Since the first experimental observation of the brain in the early 20th century researchers has observed that the amount of activity of the brain at rest is nearly the same as during an active task.

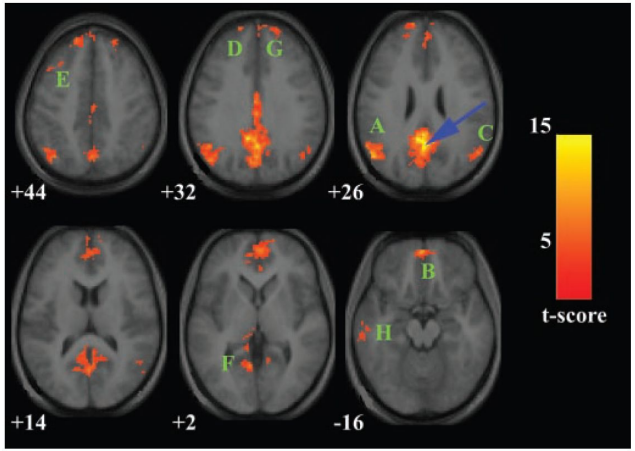
In 1997, Shulman et al. observed the existence of *task-negative network* that seemed to be deactivate during active tasks, like language, motor or visual goal-directed tasks.



The Default-Mode Network

In 2001, Raichle et al. observed the same pattern of activated brain areas in subjects at rest using PET imaging. They coined the term *Default Mode Network* for this particular network.

In 2003, Greicius et al. discover the same network with fMRI resting state data.



The Default-Mode Network

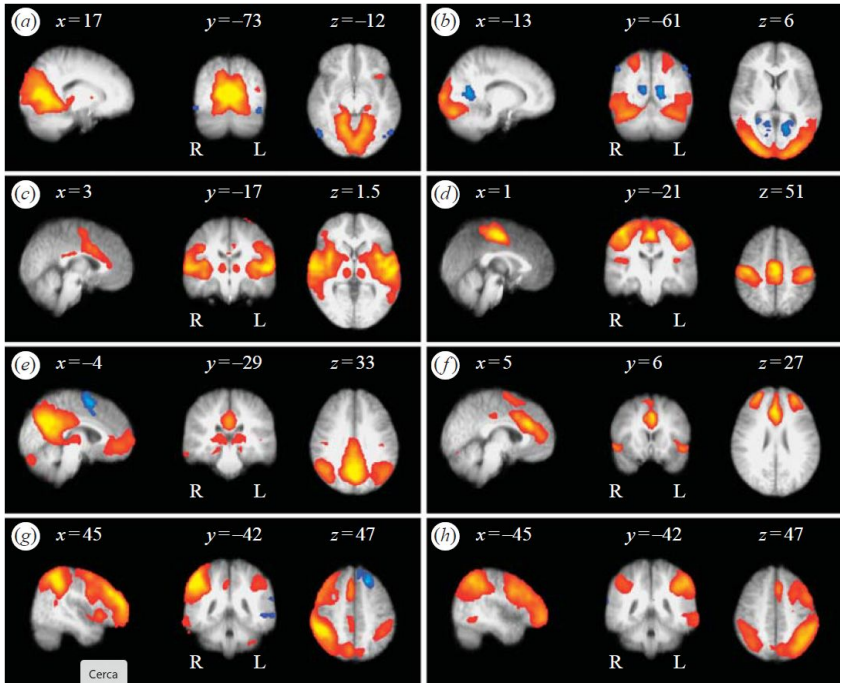
Following this initial works, many studies have addressed the role of the Default Mode Network with respect to representation of the self, thinking about others and remembering the past.

Alteration of the DMN have also been related to clinical disorders such as Alzheimer's disease, autism depression and many others.

The brain region composing the DMN are the medial prefrontal cortex, the posterior cingulate cortex, the precuneus and the angular gyrus.

The Resting State Networks

In 2005, Beckmann et al. Defined a set of resting-state networks performing an ICA analysis over resting-state fMRI data.



Our lectures about functional connectivity

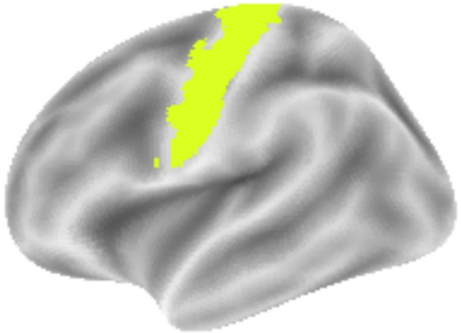
01 - Seed-based FC

02 - Graph analysis applied to resting-state fMRI

03 - Dynamic Functional Connectivity

01 - Seed-based FC

Just choose a seed region of the brain



And then correlate the average time series of the seed with all the voxels of the brain

It’s simple :)

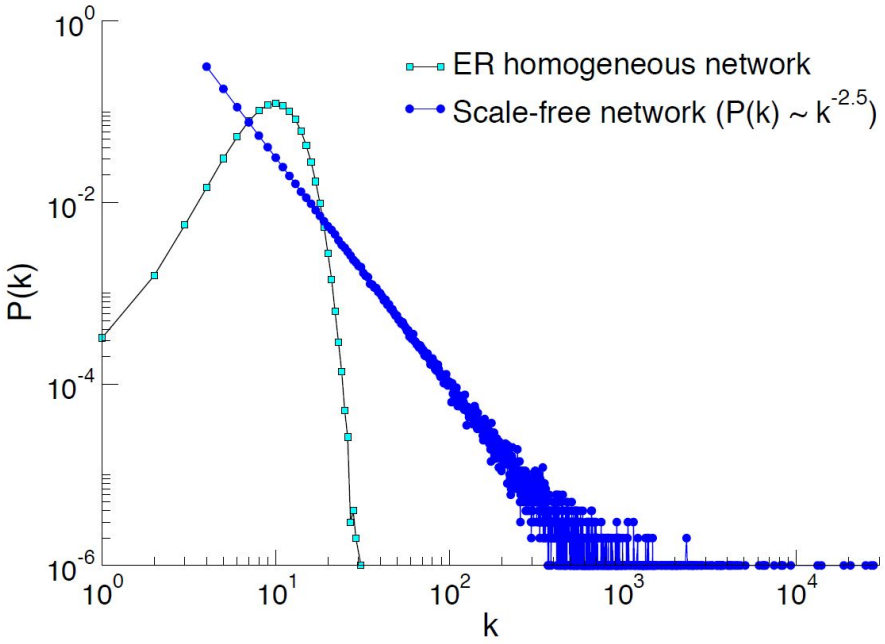
02 - Graph analysis applied to resting-state fMRI

Definitions

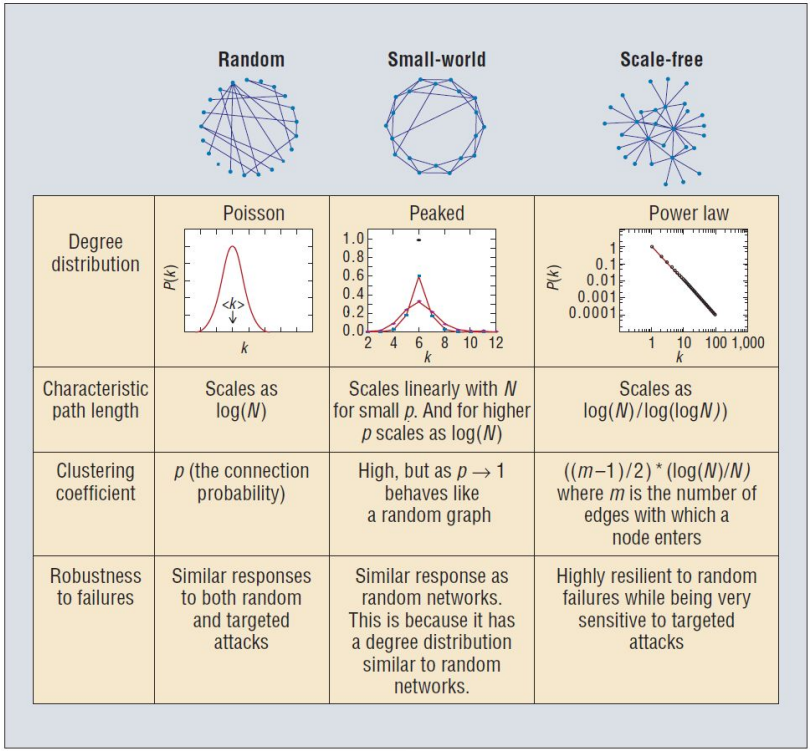
$$G = (V, E) : Graph$$
$$V = \{v_1, \ldots, v_n\} : Set\ of\ vertices$$
$$E = \{\{v_1, v_2\}, \ldots, \{v_n, v_m\}\} : Set\ of\ edges$$
$$A = (a_{ij}) : Adjacency\ matrix,$$
$$where\ a_{ij} = \begin{cases} 0 & if\ \{v_i, v_j\} \in E, \\ 1 & if\ \{v_i, v_j\} \notin E \end{cases}$$

02 - Graph analysis applied to resting-state fMRI

Random graphs, scale-free property and small wordness



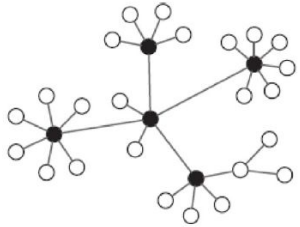
Boronchelli et al., 2013



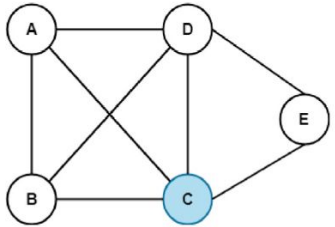
Thadakamalla et al., 2004

02 - Graph analysis applied to resting-state fMRI

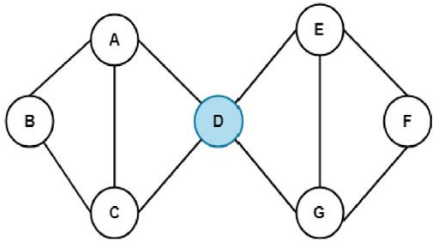
Centrality measures



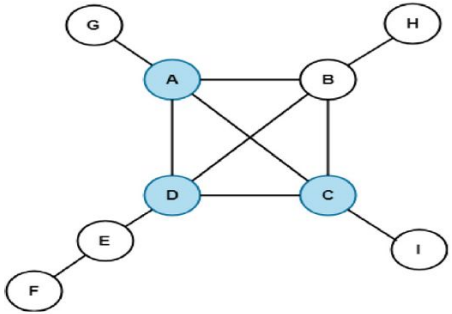
(A) Degree Centrality



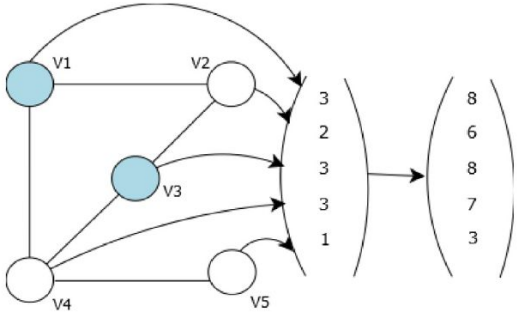
(B) Closeness Centrality



(C) Betweenness Centrality



(D). Jordan Centrality

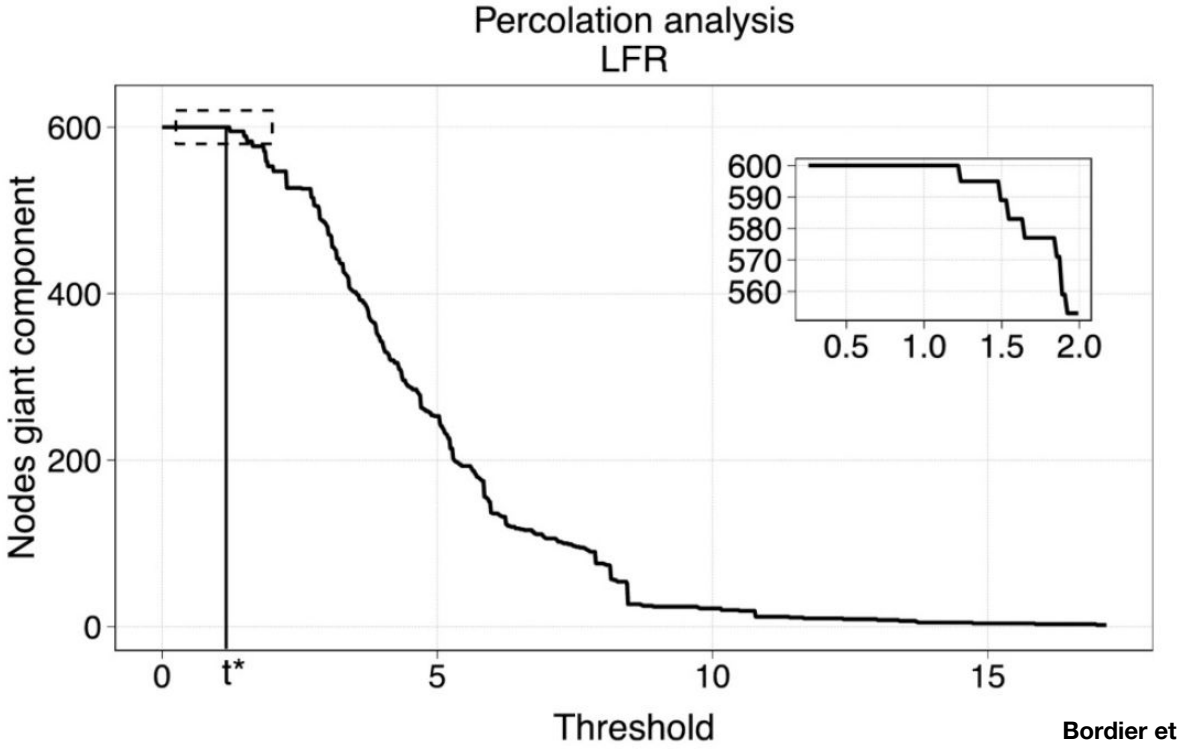


(E) Eigenvector Centrality

Shelke et al., 2019

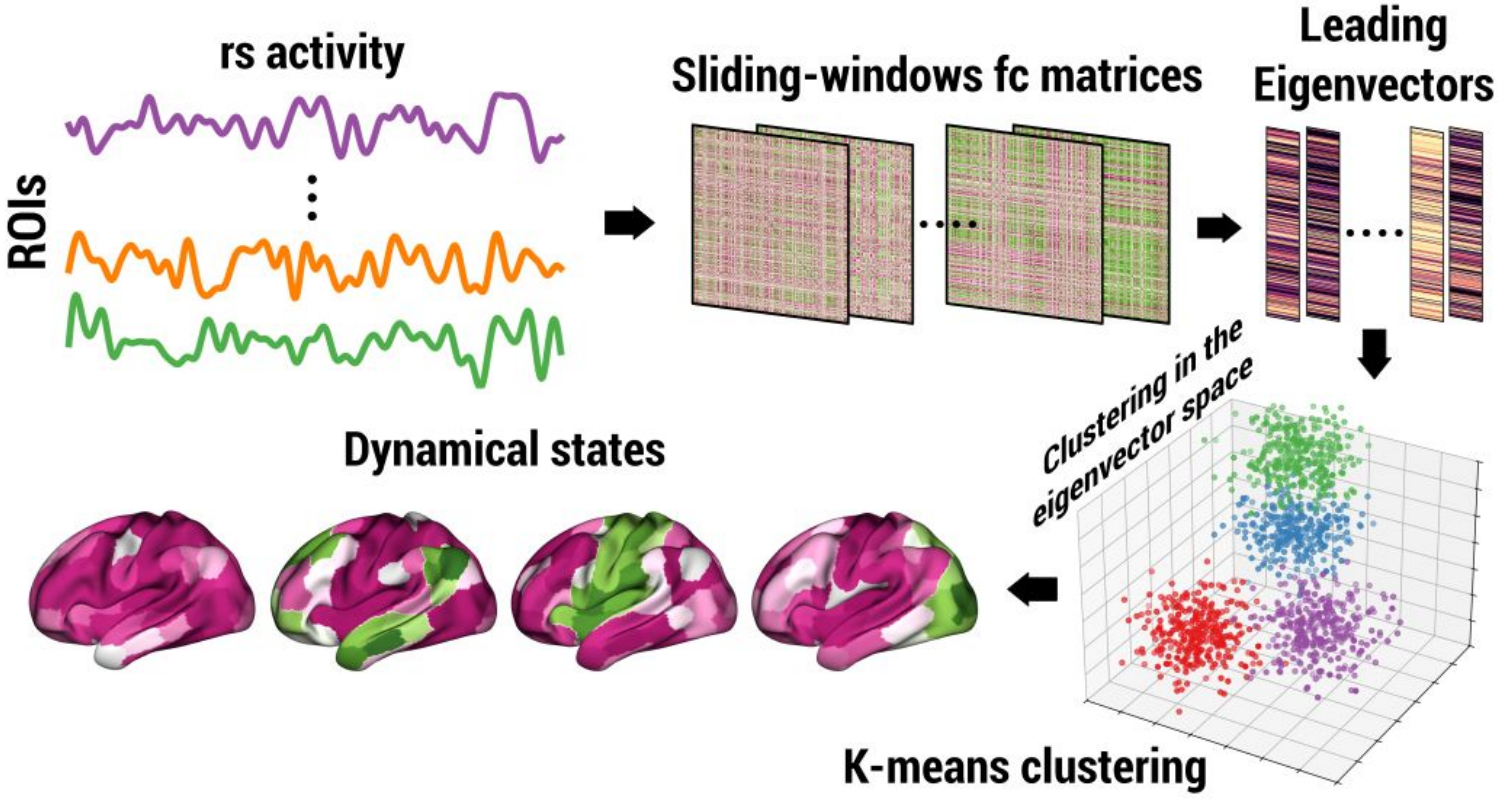
02 - Graph analysis applied to resting-state fMRI

Percolation



Bordier et al., 2013

03 - Dynamic Functional Connectivity



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